

Brontes: A Synthesis-First Wave U-Net Enhancer for Neural Codec Repair and Bandwidth Extension

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Abstract

This report presents **Brontes**, a waveform-domain enhancement model designed to repair neural codec artifacts while upsampling from 24 kHz degraded speech to 48 kHz target audio. Initial experiments used a conventional Wave U-Net with adversarial training, but repeatedly encountered the same failure mode: the model improved inputs only marginally and tended to preserve codec artifacts.

The central architectural insight was that dense skip connectivity, while beneficial for separation tasks, becomes counterproductive when the input's fine details are themselves degraded. Brontes addresses this through a synthesis-first design: selective deep skips only, aggressive temporal compression, an LSTM bottleneck, transposed-convolution decoding, gated decoder fusion, and an adversarial stack emphasizing perceptual realism in both time and spectral domains

This report explains the problem framing, the reasoning behind each design decision, the discriminator representation changes that stabilized training, and the observed qualitative outcomes.

1. Introduction

Wave U-Net architectures succeed in audio tasks by combining multiscale context with direct information transfer via skip connections. In separation tasks such as stem extraction, the useful details are already present in the mixture. The network's job is to disentangle and route them. In this setting, skip connections are ideal because they preserve trustworthy fine structure.

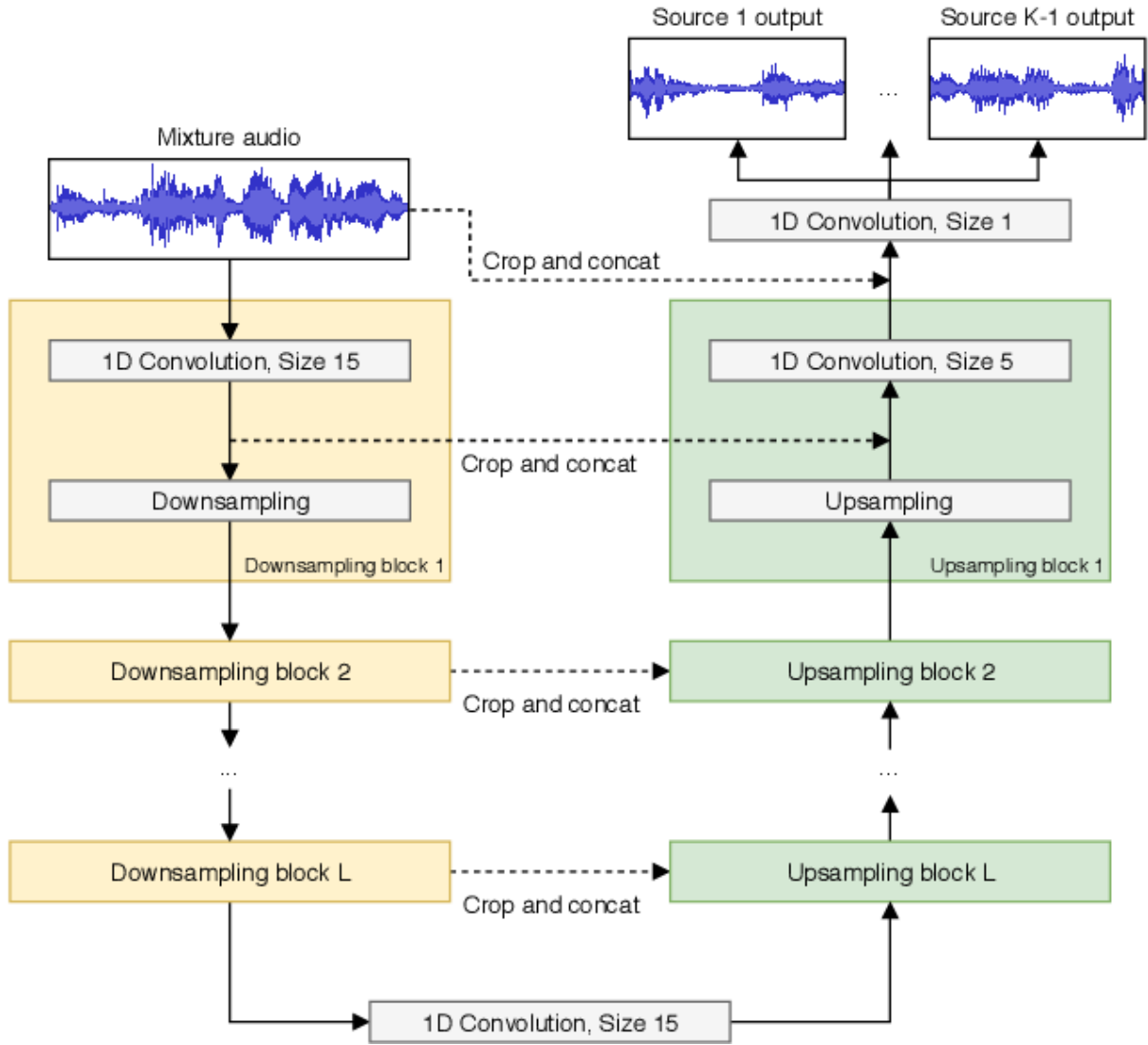


Figure 1: The standard Wave U-Net, with full connectivity

The present task is different. Here, inputs are not clean mixtures, but signals already degraded by neural codec compression. The training objective is therefore not only to reconstruct and upsample but also to actively remove compression artifacts and correct distribution mismatch. In this regime, the model should not simply pass through local detail from the input, because those details may be part of the error we are trying to remove.

Brontes emerged from this mismatch between inherited U-Net assumptions and the actual requirements of codec repair. The model retains the strengths of waveform-domain multiscale processing but constrains the information pathways that encourage copying. The resulting architecture forces constructive synthesis where it matters, while retaining enough global conditioning to preserve speaker and content identity.

2. Task Formulation and Data

The supervised task is paired audio transformation. Each sample contains a degraded input waveform and a corresponding higher-fidelity target waveform. Inputs come from speech after neural codec processing at 24 kHz; targets are original 48 kHz references. The model learns a mapping from degraded low-bandwidth speech to repaired full-band speech.

The working dataset comprises approximately 175,000 paired files of confidential multi-speaker audio. During training, paired chunks are sampled synchronously so that both input and target refer to the same temporal region. In the 48 kHz configuration, chunk length is 16,384 samples: enough local context for artifact correction while keeping throughput practical on a single accelerator. The dataset includes diverse speakers and recording conditions, which proved important for generalization.

3. Why the Initial Wave U-Net Configuration Failed

Early experiments used a conventional Wave U-Net with HiFi-GAN-style adversarial training. Although outputs often sounded marginally better than degraded inputs, improvement was small and characteristic artifacts persisted. Multiple tuning rounds varied reconstruction losses, perceptual losses, GAN loss variants, normalization modes, and channel schedules, all with no success.

The root cause was architectural, not optimization related. **Dense skips gave the decoder an easy path to simply reproduce degraded structure from the encoder.** With such a strong shortcut available, the network faced insufficient pressure to invent cleaner high-frequency detail or correct artifact patterns. **The model learned conservative refinement when the task demanded active reconstruction.**

This diagnosis reframed the project: instead of tuning losses around the same topology, the topology itself had to change so that synthesis became the default behavior.

4. Brontes Generator Architecture

Brontes is a waveform U-Net refiner with Demucs-inspired design choices, adapted for restoration rather than separation. In the 48 kHz profile, the encoder uses six stages with stride 4 per stage, producing an aggregate temporal compression of $4^6 = 4096\times$. The channel progression follows `base_ch=64` with multipliers `[2, 3, 4, 6, 6, 8]`, yielding a deep latent representation with broad receptive field. The model totals approximately 29M parameters.

At the bottleneck, Brontes replaces purely convolutional processing with a two-layer LSTM. This improves long-range temporal modeling at the bottleneck, where local convolutions alone cannot capture sufficient context. Decoder stages use transposed convolutions for learned upsampling and include GLU-style gating, encouraging controlled information flow during reconstruction.

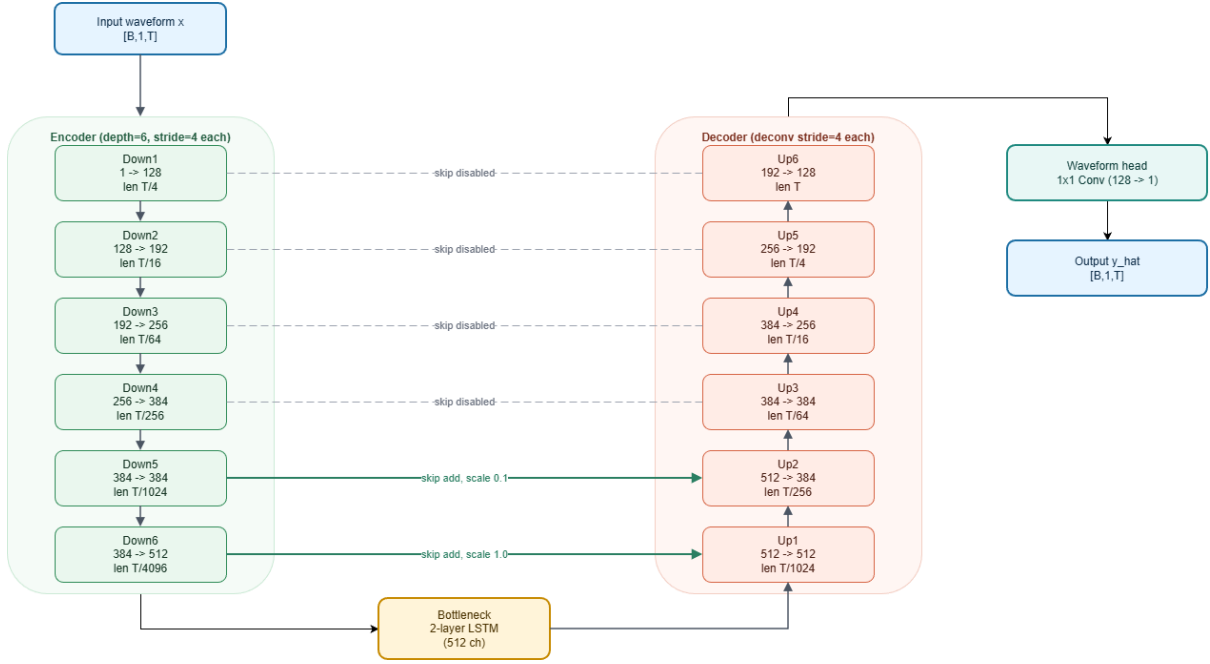


Figure 2: Brontes architecture (48KHz config) - drawn by GPT 5.3 Codex

The most consequential design choice is skip policy. Brontes does not use full skip connectivity. It retains only the two deepest encoder-to-decoder skips ([-1, -2]) with residual scales [1.0, 0.1]. This gives the decoder access to high-level semantic guidance while suppressing direct transfer of low-level degraded details. Shallower skips were observed to reintroduce artifact leakage and highly degrade quality.

We tried an iSTFT head, but that cause stability issues. The model therefore remains fully time-domain in its generator path, predicting repaired waveform samples directly.

5. Adversarial Setup and Objective Design

Brontes training combines paired reconstruction/perceptual supervision with adversarial learning. The generator objective is:

$$\mathcal{L}_G = \mathcal{L}_{audio} + \lambda_{adv} \mathcal{L}_{adv} + \lambda_{fm} \mathcal{L}_{fm}$$

where $\mathcal{L}_{audio}$ includes multi-scale mel supervision and pitch-aware loss (MR-STFT loss is disabled in this profile). Waveform L1 loss is optionally available but disabled by default.

The discriminator side copies the Descript Audio Codec recipe: Multi-Period Discriminator (MPD) and multi-band spectral discriminator. Hinge loss is used throughout.

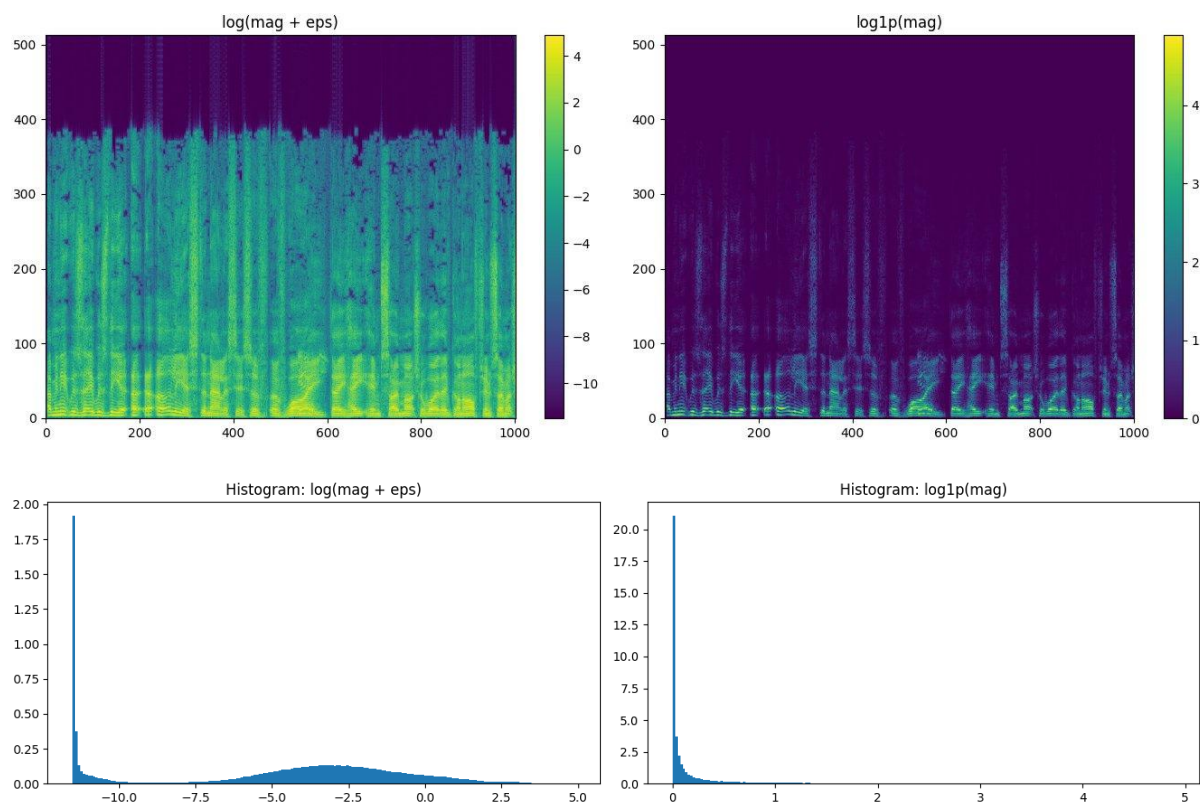
To stabilize early learning, training begins with a generator-only pretrain phase of 10,000 steps before full GAN training begins. After this transition, the generator's optimizer is reset, and both update jointly. This schedule gives the generator a viable baseline before adversarial pressure increases, reducing early collapse risk in a difficult restoration task.

6. Modified Spectral Discriminator Representation

A critical improvement came from changing how spectral discriminators see the signal. Instead of raw complex STFT channels, the modified representation uses three channels per time-frequency bin:

1. **$\log(1 + |X|)$ magnitude**: preserves low-energy structure better than plain log
2. **$\sin(\phi)$** : Sine of phase
3. **$\cos(\phi)$** : Cosine of phase

The rationale is twofold: gradient stability and representational geometry. Raw phase angles wrap discontinuously at $\pm\pi$, which creates optimization difficulties. A sin/cos embedding instead provides a smooth periodic representation. For magnitude, $\log_{1p}$ preserves perceptually relevant low-energy structure better than vanilla log in this workflow.



Empirically, this representation stabilized G-D balance and prevented the discriminator from wasting capacity on trivial low-energy bins. Alternative configurations failed decisively: raw magnitude/phase caused training collapse around 5k adversarial steps; discriminating raw complex STFT produced NaN losses almost immediately.

An additional phase gate masks phase contribution when local energy falls below a threshold. This prevents the discriminator from over-indexing on essentially uninformative phase behavior in silence or near-silence, further improving optimization focus.

7. Training Profile and Outcomes

The reference run was trained at 48 kHz with 16,384-sample chunks and AdamW optimization ($\text{lr}=1\text{e-}4$, $\text{betas}=(0.8, 0.99)$) under BF16 mixed precision. The adversarial setup used hinge loss, MPD plus multi-band spectral discrimination, and feature matching with weight 1.5. Gradient clipping is configured separately for generator and discriminator.

Wall-clock training for the successful configuration was roughly three days on a single AMD Instinct MI300X using PyTorch/ROCm.

Under the synthesis-first architecture and modified discriminator representation, training progressed qualitatively differently from earlier attempts. The generator ceased behaving as a near-identity passthrough. Instead, it reconstructed higher-frequency structure, repaired codec artifacts, and produced outputs better matching the target character. Importantly, the model generalizes well across speakers not seen during training.

Because of our limited compute and time, we could not run ablations.

Also, due to minimal filtering, the dataset is suboptimal, and model performance isn't as good as it could be. However, fine-tuning on a more specific dataset for a production model is easy and bound to deliver much better results.

8. Conclusion

The broader lesson is that architecture priors must match task semantics. **U-Net skips embody the prior that input detail is trustworthy and worth preserving.** This prior holds for separation and denoising but becomes counterproductive when detail is itself corrupted by codec artifacts. In such cases, forcing a stronger synthesis pathway unlocks behavior that loss tuning alone cannot achieve.

Brontes represents a controlled shift along this spectrum: still multiscale and still efficient, but less permissive in direct detail transfer and more explicit about generative reconstruction. The discriminator-side changes follow the same philosophy by presenting gradient-friendly spectral targets that emphasize meaningful differences.

We demonstrate that waveform enhancement quality for codec repair improves substantially when the model is deliberately pushed away from copy-heavy U-Net behavior toward synthesis-first reconstruction. We combine selective deep skips, a Demucs-style generative decoder, LSTM bottlenecking, and representation-aware spectral discrimination. Together, these changes shift training dynamics from persistent under-correction toward effective artifact repair and bandwidth extension.

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